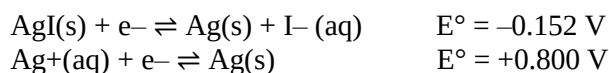


CHE 2108 Physical Chemistry Laboratory.
Revision Quiz 02

- 1) What do you understand by a eutectic point?
- 2) How do you find the eutectic point on a phase diagram?
- 3) Describe Biphenyl-Naphthalene solid –solid system.
- 4) Describe Phenol-water system.
- 5) At 298 K, the EMF of the cell shown below is +0.84 V.

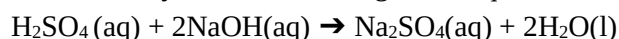


- (i) Define what is meant by the standard EMF of the cell.
 - (ii) Write down the cell reaction and the Nernst equation for the cell.
 - (iii) Calculate the equilibrium constant for the cell reaction at 298 K.
- 6) From the following standard electrode potential data, calculate the solubility product (K_{sp}) of AgI at 298 K.



- 7) The solution in contains 10.0 g of cobalt(II) chloride dihydrate, $\text{CoCl}_2 \cdot 2\text{H}_2\text{O}$, in enough ethanol to make exactly 500 mL of solution. What is the molar concentration of $\text{CoCl}_2 \cdot 2\text{H}_2\text{O}$? (Hint= $\text{CoCl}_2 \cdot 2\text{H}_2\text{O}$ is 165.87 g/mol.
- 8) An experiment required 200.0 mL of a 0.330 M solution of Na_2CrO_4 . A stock solution of Na_2CrO_4 containing 20.0% solute by mass with a density of 1.19 g/cm³ was used to prepare this solution. Describe how to prepare 200.0 mL of a 0.330 M solution of Na_2CrO_4 using the stock solution.
- 9) Use the information to determine the concentration of the sulfuric acid.
- A 25 cm³ sample of sulfuric acid is sucked into a pipette and transferred into a 500 cm³ volumetric flask. The solution is made up to the mark.
 - 25 cm³ of the diluted acid is transferred into a conical flask using a pipette.
 - A burette is used to neutralise the acid with 0.1 mol dm⁻³ sodium hydroxide.

Sulfuric acid reacts with sodium hydroxide according to the equation:



- (i) The average titre of the sodium hydroxide solution was 25.00 cm³. Calculate the number of moles in the average titre.
 - (ii) Determine the number of moles in the diluted sample of sulfuric acid, and hence the concentration of the diluted acid.
 - (iii) Calculate the concentration of the undiluted sulfuric acid in mol dm⁻³.
 - (iv) Calculate the concentration of the sulfuric acid in g dm⁻³.
- 10) What is an equivalence point?